IOT LABS - PYTHON ASSIGNMENT (DICTIONARY + SETS)

IOT Labs - Assignment Part 1 (Dictionary)

Q1: Count occurrences without count()
Input: [2, 3, 2, 5, 3, 2, 5, 8] -> Output: {2:3, 3:2, 5:2, 8:1}
Solution:

```
numbers = [2, 3, 2, 5, 3, 2, 5, 8]
result = {}
for num in numbers:
    if num in result: result[num]+=1
    else: result[num]=1
print(result)
```

Q2: marks = {'Ali':85, 'Sara':95, 'Ahmed':95, 'Noor':78}
Print toppers.
Solution:

```
marks = {'Ali':85,'Sara':95,'Ahmed':95,'Noor':78}
highest = max(marks.values())
for name, score in marks.items():
    if score==highest: print(f"{name}: {score}")
```

Q3: data = {'a':1,'b':2,'c':3} swap keys/values
Solution:

```
data={'a':1,'b':2,'c':3}
swapped={}
for k,v in data.items(): swapped[v]=k
print(swapped) -> {1:'a',2:'b',3:'c'}
```

Q4: words = ['apple','ant','banana','ball','cat']
Group by first letter.
Solution:

```
words=['apple','ant','banana','ball','cat']
grouped={}
for word in words:
    fl=word[0]
    if fl not in grouped: grouped[fl]=[]
    grouped[fl].append(word)
print(grouped)
```